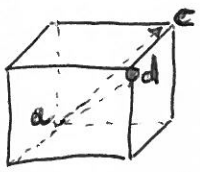
P1



$$\vec{ac} = \vec{C} = (0,1,1) = \hat{j} + \hat{k}$$
$$\vec{ad} = \vec{D} = (1,1,1) = \hat{i} + \hat{j} + \hat{k}$$

(a) $\vec{C}\cdot\vec{D} = 2 = |\vec{C}||\vec{D}|\cos\varphi = \sqrt{2}\cdot\sqrt{3}\cos\varphi$

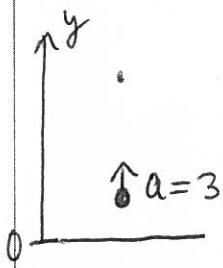
$$\cos\varphi = \frac{2}{\sqrt{2\cdot 3}} = \sqrt{\frac{2}{3}} \Rightarrow \varphi = 35.2^\circ$$

(b) by definition $\vec{A} = \vec{C}\times\vec{D} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \hat{i}(1-1) - \hat{j}(0-1) + \hat{k}(-1)$

$$= \hat{j} - \hat{k} = (0,1,-1)$$

Note that $\vec{D}\times\vec{C} = -\vec{A}$ is also a solution.

P2



$a = 3.50\,m/s^2$ for $t \le t_1 = 10.0\,s$

(a) $y(t_1) = \underset{0}{y(0)} + \underset{0}{v_0 t_1} + \frac{1}{2}at^2 = \frac{1}{2}\,3.50\,(10.0)^2 = 175\,m$

$$v(t_1) = \underset{0}{v_0} + at_1 = 35.0\,m/s$$

(b) For times $t > t_1$ we have rocket in a free-fall condition: $v(t_2) = v(t_1) - gt_2 = 0$ at the point of maximum height

$$\Rightarrow t_2 = \frac{35\,m/s}{9.80\,m/s^2} = 3.57\,s$$

vertical displacement $\Delta y = v(t_1)t_2 - \frac{1}{2}gt_2^2 = 62.5\,m$

Maximum height $= 175 + 62.5 = 237.5\,m$

time to get there $= 10 + 3.57 = 13.6\,s$

P3 (a) The one with greater velocity is ahead

$$V_A(t) = \frac{dx_A(t)}{dt} = \alpha + 2\beta t$$

$$V_B(t) = \frac{dx_B}{dt} = 2\gamma t - 3\delta t^2$$

$V_A(t=0) = \alpha$ but $V_B(t=0) = 0$

$\Rightarrow$ A is ahead

(b) $x_A(t) = x_B(t)$

$\alpha t + \beta t^2 = \gamma t^2 - \delta t^3$

$t=0$ (initial position of both is at zero) or

$\alpha + \beta t = \gamma t - \delta t^2$

$\delta t^2 + (\beta - \gamma)t + \alpha = 0$

$$t = \frac{\gamma - \beta \pm \sqrt{(\gamma-\beta)^2 - 4\alpha\delta}}{2\delta} =$$

$$= \frac{1.6 \pm \sqrt{1.6^2 - 4 \cdot 2.6 \cdot 0.2}}{0.4} = \begin{cases} 5.725 \text{ s} \\ 2.275 \text{ s} \end{cases}$$

Thus, $t=0$, $t = 2.27$s, $t = 5.72$s are solutions.

(c) $a_A = \frac{dV_A}{dt} = 2\beta$

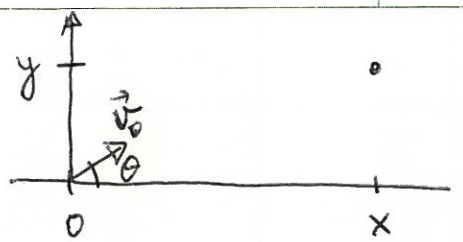
$a_B = \frac{dV_B}{dt} = 2\gamma - 6\delta t$

$2\beta = 2\gamma - 6\delta t$

$6\delta t = 2(\gamma - \beta)$

$t = \frac{1}{3}\frac{\gamma - \beta}{\delta} = 2.67$s

P4



$x = 11.0\,m$, $y = 3.05\,m$

$$y = v_{0y} t - \frac{1}{2} g t^2$$

$$x = v_{0x} t \Rightarrow t = \frac{x}{v_{0x}} = \frac{x}{v_0 \cos\theta}$$

$$y = v_0 \sin\theta \frac{x}{v_0 \cos\theta} - \frac{g}{2} \frac{x^2}{v_0^2 \cos^2\theta} = x \tan\theta - \frac{g x^2}{2 v_0^2 \cos^2\theta}$$

(a) Minimum angle θ_{min} is such that (corresponds to $v_0 = \infty$)

$$\tan\theta_{min} = \frac{y}{x} \Rightarrow \theta_{min} = 15.5^\circ$$

(b) $\theta = 45^\circ \Rightarrow \tan\theta = 1$, $\cos^2\theta = \frac{1}{2}$

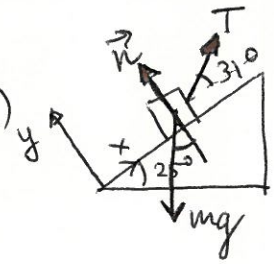
$$\Rightarrow y = x - \frac{g x^2}{v_0^2} , \quad \frac{g x^2}{v_0^2} = x - y$$

$$v_0^2 = \frac{g x^2}{x - y}$$

$$v_0 = \sqrt{\frac{9.8\,(11)^2}{11 - 3.05}} = 12.2\,m/s$$

P5

(a)



(b) y-axis: $n - mg\cos 25^\circ + T \sin 31^\circ = 0$

x-axis: $T\cos 31^\circ - mg \sin 25^\circ = 0$

$$\Rightarrow T = \frac{mg \sin 25^\circ}{\cos 31^\circ} = 5460\,N$$

(c) $n = mg\cos 25^\circ - T \sin 31^\circ = 7224\,N$

(d) $a = g \sin 25^\circ = 4.14\,m/s^2$